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Paleomagnetism of Devonian Units from Miguasha, Gaspe, P.Q., Canada

Maurice K. SEGUIN and Daniel ST-GELAIS*

Paleomagnetic Laboratory, Faculty of Sciences and Engineering, University Laval, Quebec, Canada G1K 7P4

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The area investigated is located at longitude 66° 30'W and latitude 48° N, on the northern shore of Chaleur Bay. Some 124 specimens (46 samples) were collected from 9 sites in the following geological formations: La Garde (10 samples, late Early to early Middle Devonian), Pirate Cove (11 samples, early Middle Devonian), Fleurant (15 samples, early Late Devonian) and Escuminac (10 samples, early Late Devonian). These formations are composed of conglomerates, silty and calcareous sandstones deposited in a continental environment. The Upper Devonian geological units are amongst a few found in Canada to the east of Alberta. The main goal of this study is a better definition of the Apparent Polar Wander Path of the Appalachians of North America. This study allowed the identification of geological formations favourable to paleomagnetic sampling. Remanent magnetizations and magnetic susceptibilities were measured; both alternating field and thermal demagnetizations were carried out. After demagnetization, 4 components (A, B, C and D) were isolated and several components were found in each formation. Fold and conglomerate tests were performed; the first is significantly negative except for component B and the second is positive in the Fleurant Formation. The mean direction of the components is: 156°, +31° for A, 52°, −46° for B, 167°, −21° for C and 223°, +25° for D. The corresponding paleopoles are 22° N 141° E, 6° S 77° E, 51° N 137° E and 17° N 83° E. Based on the fold test, degree of alteration and unblocking temperatures, it was found that components A, C and D are secondary. Component B is either primary or secondary. The relative age of acquisition of magnetization is the following: $B > D > A > C$. The paleolatitudes vary from 17° S to 10° N going from A to B (tilt corrected), to D and to C. The results are discussed in relation to the Devonian Apparent Polar Wander Path of North America.

1. Introduction

The Cenozoic and Mesozoic paleomagnetic record of North America (NA) is now relatively well established (e.g., HARRISON and LINDH, 1982; IRVING and IRVING, 1982). However, the paleomagnetic record available for the Upper Paleozoic period (Carboniferous and Permian) is still incomplete. IRVING and IRVING (1982),

*Now at Petro-Canada, 40 Research Place NW, Calgary, Alta., Canada T2L 1Y6.

ROY *et al.* (1983), and SEGUIN and FYFFE (1986) have compiled a probable reliable paleomagnetic data bank for the NA Carboniferous and Permian periods.

For ages older than Carboniferous, the APWP for NA is still ill-defined (SEGUIN and GAHÉ, 1985; IRVING and STRONG, 1985). Recent literature on Devonian paleomagnetic results of North America (DANKERS, 1982; JOHNSON and VAN DER VOO, 1983; SPARIOSU and KENT, 1983; ROY *et al.*, 1983; SEGUIN, 1986) show that the paleomagnetic data set is either too limited in number and/or suffers from too many ambiguities to make reliable Paleozoic continental restorations. For instance, VAN DER VOO (1981) concluded that there was a small northward oriented Apparent Polar Wander Path (APWP) along longitude 105° E for the Devonian period while SEGUIN (1986) arrived at a loop centered at latitude 10° N and bounded by longitudes 90° and 145° E. The first interpretation implies little polar motion and the second, a large one. The amount of scatter in published pole positions for the Devonian period seems to preclude any discussion of the APWP or of relative motion of displaced terrains during the Acadian orogeny. The construction of the Devonian APW path of NA may become better constrained as new and reliable paleomagnetic data are published. The present paleomagnetic study from Middle and Upper Devonian rock units from the Miguasha area is a contribution towards this objective.

2. Location and Geological Setting

The area investigated is located at the western extremity of Chaleur Bay on the south of the Gaspé Peninsula just north of New Brunswick at longitude 66° 33'W and latitude 48° 15'N. The local geology is described by ALCOCK (1935) and DINELEY and WILLIAMS (1968a, b). The region sampled is part of a large sedimentary basin of Devonian age covering a surface of approximately 1500 km^2 (Fig. 1). The Matapedia Group of Upper Ordovician age constitutes the base of this basin. Sediments of the Silurian Chaleurs Bay Group conformably to unconformably overlie the Matapedia Group (BOURQUE, 1975). Sedimentary and volcanic rock units of Devonian age conformably overlie the Chaleurs Bay Group. Volcanic rocks which are abundant in the Lower Devonian section are absent in the unconformably overlying La Garde Formation. This last formation is overlain conformably by the Pirate Cove Formation which, in turn, is unconformably overlain by the Miguasha Group (Fleurant and Escuminac formations). According to DINELEY and WILLIAMS (1968a), the Fleurant Formation is separated from the early Middle Devonian Pirate Cove Formation by an irregular unconformity but ZAITLIN (1981) states that the boundary of the overlying Escuminac Formation is abrupt and he interprets it as a disconformity. Bonaventure red conglomerate of Mississippian to Pennsylvanian age lies unconformably on top of these formations. The nine sites sampled are representative of the La Garde (2), Pirate Cove (2), Fleurant (3) and Escuminac (2) formations (Fig. 1).

A short description of each sampled formation is given below. The composition of the La Garde Formation is variable. It consists of sandstones, siltstones, shales, grey and green conglomerates with interbeds of impure limestone. At the sampled

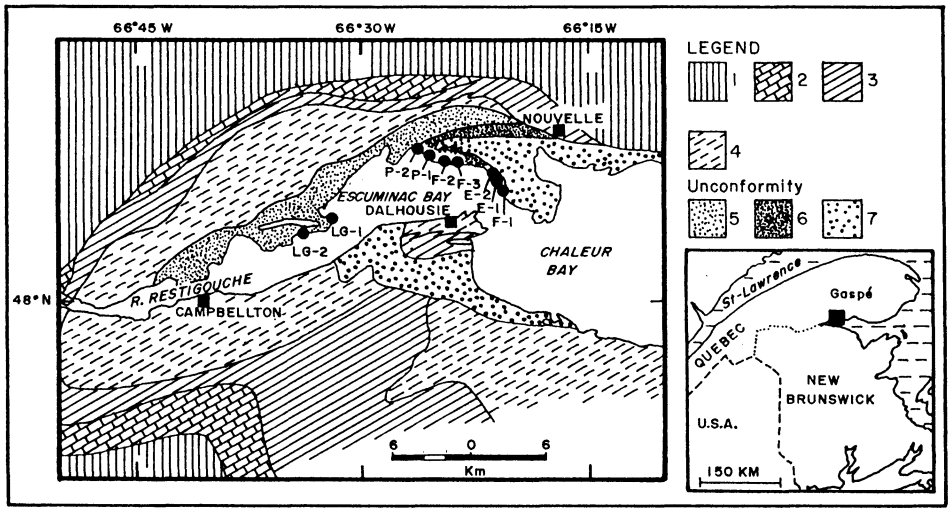


Fig. 1. Simplified geological map of the Escuminac Bay region, southern Gaspé, Quebec, with the distribution of paleomagnetic sampling sites as indicated by numbers next to solid circles in the vicinity of Miguasha, Fleurant, Pirate Cove and Escuminac. Lithology caption. 1: Matapedia Group, 2: Chaleurs Bay Group, 3: Lower Devonian volcanic rocks, 4: Lower Devonian volcano-sedimentary rocks, 5: La Garde Formation, 6: Pirate Cove Formation and Miguasha Group, 7: Bonaventure Formation (Carboniferous).

sites, limestone was absent. The age of this formation is late Early to early Middle Devonian (Emsian–Early Eifelian). The Pirate Cove Formation has an early Middle Devonian age and is composed of conglomerates containing angular clasts intercalated with beds of sandstones and red shales which are often limy. According to DINELEY and WILLIAMS (1968a), the source of these alluvial sediments is located to the northwest. The Fleurant Formation is a sequence of conglomerate containing pebbles and boulders interbedded with sandstones and shales. The pebbles are found in the conglomerate but are not made of conglomerates. It is of fluviatile origin and the erosion of the Pirate Cove Formation was the local source of some of these sediments (DINELEY and WILLIAMS, 1968b). Its age is early Late Devonian. The Escuminac Formation is constituted mainly of limy sandstones, shales, brown and green conglomerates. These sediments were deposited in a lake as evidenced by the fresh water fossilized fishes found at the universally known Miguasha locality (SEGUIN, 1976). The age of the Escuminac Formation is also early Late Devonian.

3. Paleomagnetic Sampling and Measurements

In this paleomagnetic study, 46 oriented samples (124 specimens) at 9 sites were collected from well studied and dated localities. Two sites (10 samples) were sampled

on the Escuminac Formation, 3 sites (15 samples) on the Fleurant Formation, 2 sites (11 samples) on the Pirate Cove Formation and 2 other sites (10 samples) on the La Garde Formation. The sampling was carried out with a portable diamond drill and orientations were measured with a Brunton compass. Because of good exposures, geological strikes and dips were accurately noted in order to perform a fold test and pebbles could be sampled to carry out a conglomerate test.

Magnetization (M) was measured with a digital Schonstedt spinner magnetometer (DSM-1). Two methods of demagnetization were used: 1) a Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT and 2) a GSD-1 Schonstedt alternating field demagnetizer (peak intensity=100 mT) or a demagnetizer with a maximum peak-to-peak intensity of 90 mT in 3 large mu-metal cylinders (SEGUIN, 1975) incorporating a three-axis tumbler. The estimated uncertainties in the NRM determinations are about 6% in intensity and 4° in direction. The volume susceptibility (K) was measured at room temperature in the Earth's field (H=0.05 mT), using a Cenco MS-3B susceptibility bridge.

4. Mineralogy

Some 15 mineralogical observations were done under transmitted and reflected light microscopes for all 4 geological formations investigated. The Escuminac Formation is a beige grey stratified siltstone unit. Hematite constitutes about 1% of the opaque minerals and magnetite 1.5 to 2.0%. Their grain size varies from 0.01 to 0.04 mm in diameter. The Fleurant Formation is a brownish red, slightly stratified arenite containing calcite veinlets. The matrix contains 0.1 to 0.3% of hematite and the grains 1 to 1.5% magnetite and 0.5% of titanomagnetite and ilmenite. In the conglomeratic section of this formation, the matrix contains 0.5–1% hematite and 2.5% magnetite and the grains 2–2.5% hematite and 0.5–1% magnetite. After heating, some of the hematite was transformed to magnetite; this reaction is probably due to the dissociation of CO_2 from the calcite. In the Pirate Cove Formation, the sandstone as well as three pebbles were examined. This formation is a pinkish red stratified and graded bedded rudite; its matrix contains 2.5–3% hematite and 0.1–0.5% magnetite and its grains 0.5–1% hematite and 1–1.5% magnetite. Extraformational pebbles from the epiclastic conglomerate have a matrix composed mainly of hematite and grains constituted of magnetite. In the matrix, the percentage of hematite varies from 0.15–0.3% and that of magnetite 0–0.1% while in the grains, the percentage of magnetite ranges from 2 to 3.5% and that of hematite from 0.5 to 1%. Upon heating, a little magnetite is oxydized to hematite but the reverse process is also observed occasionally. Most commonly the magnetite grains are surrounded by a thin shell of hematite. The La Garde Formation is a brownish rather massive arenite; its matrix is composed of 2.5 to 4% hematite and 0 to 1% magnetite while the grains are constituted of 0.5 to 2% magnetite and 0.5 to 3% hematite. All the magnetite grains underwent an important phase of hematitization.

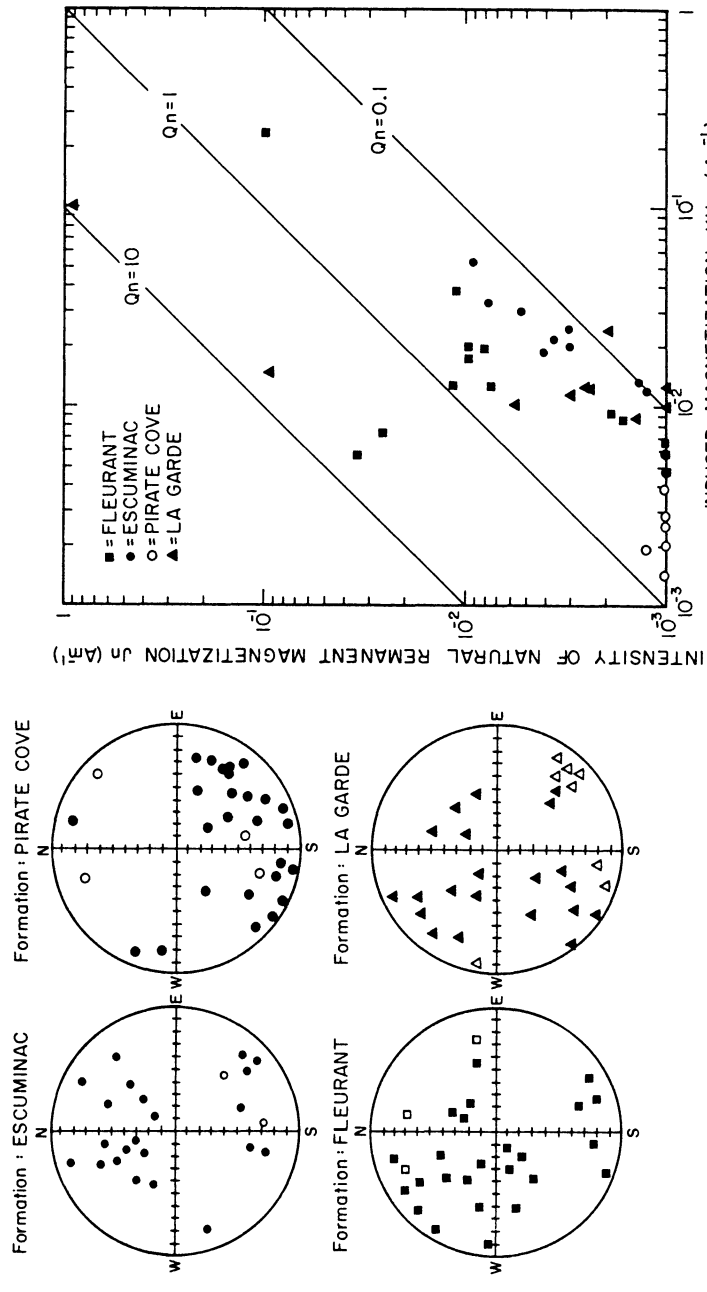


Fig. 2. Specimen NRM directions (before tilt correction) plotted on Wulff net for Escuminac, Fleurant, Pirate Cove and La Garde Formations (left). Solid (open) circles represent projections on the lower (upper) hemisphere. The star represents the present Earth's field (PEF) of the sampling region. Plots of corresponding intensity of NRM (Jn in Am^{-1}) against the induced magnetization (KH in Am^{-1}) with resulting slope lines indicating the values of Koenigsberger ratio (Qn, right).

5. NRM Directions and Intensities

Individual specimens NRM directions of the 4 geological formations, although characterized by wide scatter caused by positively and a few negatively inclined shallow to intermediate remanence and a smaller amount of steeply inclined remanence, have a pronounced deviation away from the Present Earth's Field (PEF) ($D=335°$, $I=75°$) in situ (Fig. 2). This indicates that the 4 formations do carry some ancient remanence especially since the directions of remanence of the few specimens ($\leqslant 15\%$) that lie close to the PEF were found to move away from these initial directions during AF or thermal demagnetization. The majority of the NRM directions are predominantly situated in the SE and NW quadrants on the stereonet. The intensities of NRM (Jn) vary from 10^{-3} and 10^{-2} Am^{-1} with values of Koenigsberger's ratio ($Qn=Jn/KH$) between 0.1 and 10, with most ratios between 0.2 and 0.7 (Fig. 2). The high Jn and KH values correspond to boulders of volcanic nature in the Fleurant and La Garde formations (green conglomerates).

6. Demagnetization Studies

6.1 Alternating field (AF) demagnetization

Only 47 of the 124 specimens, that is one specimen per oriented sample, were demagnetized with an alternating field. Subsequent pilot tests making use of thermal demagnetization indicated that thermal demagnetization was more efficient than AF demagnetization. So, approximately 50% of the AF demagnetized specimens were subsequently thermally demagnetized. In most cases, thermal demagnetization appeared to be more efficient than AF demagnetization. AF demagnetization was carried out by steps of 5 mT from 0 to 20 mT and steps of 10 or 20 mT from 20 to 100 mT. The median destructive field (MDF) varies from 5 to 40 mT (mean: 15) with one exception at 80 mT. The PEF is present in 10 of the 45 specimens and no other component cold be isolated besides the PEF in 5 specimens either because of magnetic instability or a residual intensity of magnetization only slightly above the instrumental noise level. Four components of magnetization (A, B, C and D) were isolated (Fig. 3). The principle of grouping and identification of the components is the following. Magnetic components are present when a specimen, sample or site contains multiple magnetization with a different direction and origin. The characteristic directions of the differing components of magnetization were identified with the help of ZIJDERVELD (1967) diagrams and subsequent statistical analysis while their origin was determined with the aid of their specific magnetic properties (coercivity spectrum, unblocking temperature range) and geological criteria (magnetic mineralogy, degree of alteration, age of orogenic events, etc.). In this investigation, only one component was detected in a specific sample but two or three components were generally found in a given site. Reversals are not frequent; they were found in 3 specimens only. In this instance a geomagnetic reversal is not observed within a specific specimen and a magnetic directional reversal has not taken place in the course

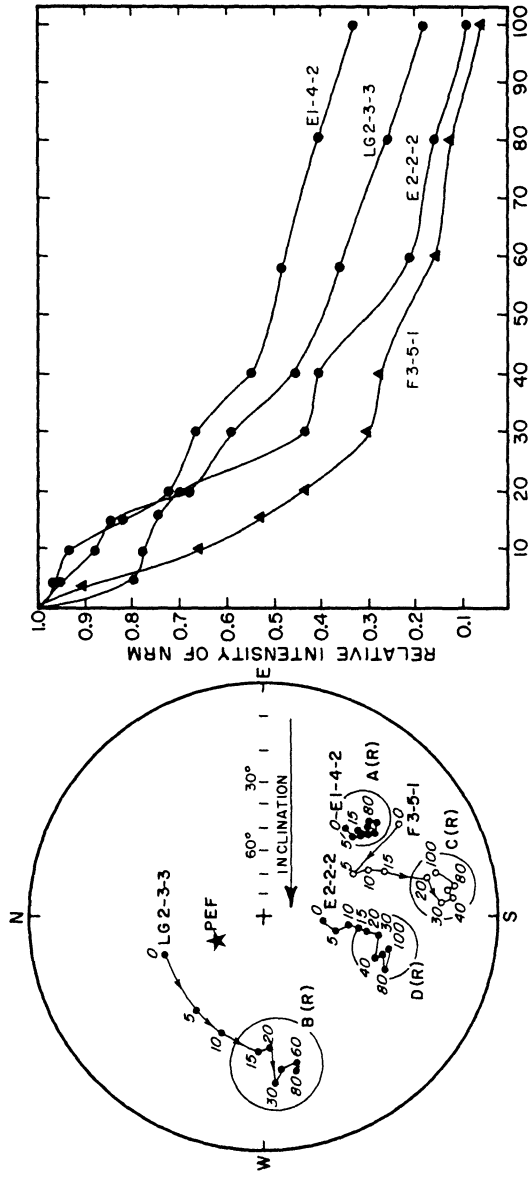


Fig. 3. Equal-area projections of directions of magnetization (left) of 4 specimens after stepwise AF treatment. The numbers next to the points represent the demagnetization steps in mT as indicated. The respective normalized AF decay curves are shown to the right.

of stepwise demagnetization. What is meant here is that some specimens recorded the reversed polarisation of a specific component which cannot represent only a very short time span. The Escuminac Formation contains components A, C and D, the Fleurant Formation components, A, C and D, the Pirate Cove and the La Garde formations components A, B, C and D.

6.2 Thermal Demagnetization

The remaining 77 specimens which include all the samples (excluding those which were AF demagnetized) were thermally demagnetized in the 20–650° C range at the following temperatures: 100, 200, 300, 400, 450, 500, 540, 570, 600, 625 and 650° C. The unblocking temperatures (T_{UB}) vary from 400 to 630° C. Most of the T_{UB}s are located in the 540–570° C range and 8 of them in the 610–630° C range. The PEF is present in 9 of the 77 specimens; no other components besides the PEF are observed in 4 of the specimens. Some 5 specimens are definitely unstable and 4 others are really too weak; their intensity is only of 2 orders of magnitude that of the instrumental noise. Artificial remanence expressed by an ever increasing intensity as a function of temperature and due to either reduction of hematite or oxydation of titanomagnetite is observed in 8 specimens. Four components of magnetization (A, B, C and D) were also isolated (Fig. 4). The components of magnetization, which are not identical to directional groupings, are distinguisbed on the basis of the magnetic directions, the unblocking temperature window and the geological information. Excluding the PEF, two superimposed components of magnetization were found in 12 specimens only. Most commonly, component C was extracted at a higher temperature than components A or D but this was not always the case. In the case of component B, the situation was different; this last component was always isolated at higher temperatures than components A or D. The T_{UB}s of component A range from 400 to 570° C (mean=500° C), those of component D from 400 to 625° C (mean=520° C) with 2 values at 625° C, those of component C from 350 to 625° C (mean=540° C) with 4 values in the 610–625° C range and finally those of component B from 400 to 650° C (mean=545° C) with 5 values in the 620–650° C range. Pending on the time interval of a geological event, single or dual polarisations of the paleofield may be detected; short timespans are characterized by a single polarity while reversals are generally observed on longer time intervals in a given geological formation. The spatial distribution of the reversals is a function of the geological process involved during remagnetization. Reversals are not common (10 specimens); two were found in component A, one in components C and D respectively and 6 or 7 in component B. The Escuminac Formation contains components A and C, the Fleurant Formation components A, C and D, the Pirate Cove and La Garde formations components A, B, C and D.

7. Mean Directions of Magnetization

Table 1 shows the structural data and the sample mean directions of magnetization of the in situ NRM values. In the mean of all sites ($D=341°$, $I=76°$), the generally high inclinations, the large scatter in declinations and the large α_{95} values are

suggestive of a PEF component superimposed on one or more older components. Clearly, some of the directions (e.g., a large fraction of those from the Pirate Cove Formation) are not in agreement with the mean of all sites, mainly because very few directions are characterized by a steep inclination.

The distribution of the NRM directions before demagnetization is entirely different from those after demagnetization. This demonstrates without any doubt the existence of old components of magnetization (secondary and possibly one primary). The extraction of 4 older components with the help of AF and thermal demagnetization methods necessitated the trial and use of different analytical techniques. Attempts to use the technique of the great circle path (HALLS, 1976) on some 50 specimens did not succeed. Vector subtraction techniques were useful in the 25 specimens carrying two or more components. In all other cases, orthogonal diagrams (Fig. 4) and stereonet plots (Fig. 3) were most useful. Component A has a southeast direction, a shallow to intermediate downward inclination and is characterized by a definite reverse polarity (Figs. 3 and 4(a)). This component is isolated in 7 of the 9 sites investigated and it is present in all four geological formations (Fig. 5). Component B is characterized by a ENE intermediate negative direction with a mixed and predominantly normal polarity (Figs. 3 and 4(b)). This component is not common and is isolated in only 4 of the 9 sites. Component B (Fig. 5) is present only in the two lower formations (Pirate Cove and La Garde). Component C has a mean southern shallow negative direction (Figs. 3 and 4(c)). It is isolated in 6 of the 9 sites and predominantly in the Pirate Cove, Fleurant and La Garde formations (Fig. 5). This component which is not as common as component A but more common than component B is characterized by a predominantly reverse polarity. Component D is detected in 7 sites all of which are mainly in the Fleurant, Pirate Cove and La Garde formations (Fig. 5). This component has a mean SSW orientation with a shallow downward inclination (Figs. 3 and 4(d)). The polarity of this component is clearly reversed. Table 2 summarizes the results obtained for each component.

8. Fold and Conglomerate Tests

The fold test is significantly negative for components A, C and D (MCELHINNY, 1964). In the case of component B, the fold test is positive but not significantly since the k value after tilt correction is only slightly larger than the in situ k value. This relative invariance in the k values for component B may result from the small variations in the dip of the formations (Table 1) or the insufficient cleaning of this component. It is thus concluded that components A, C and D are definitely secondary while component B may be primary or secondary. Since component B is detected only in the La Garde and Pirate Cove formations which are of late Early and early Middle Devonian age, it is quite certain that this component is older than early Late Devonian. Indeed, there is no specific reason why it is not found in the Fleurant and Escuminac formations as well. The other argument is the following: the paleopole of the cleaned in situ direction (52°, −55°) for component B is 75° E, 6° S. A look at the expected APWP for the Devonian period (Fig. 6) shows that the pole position of in

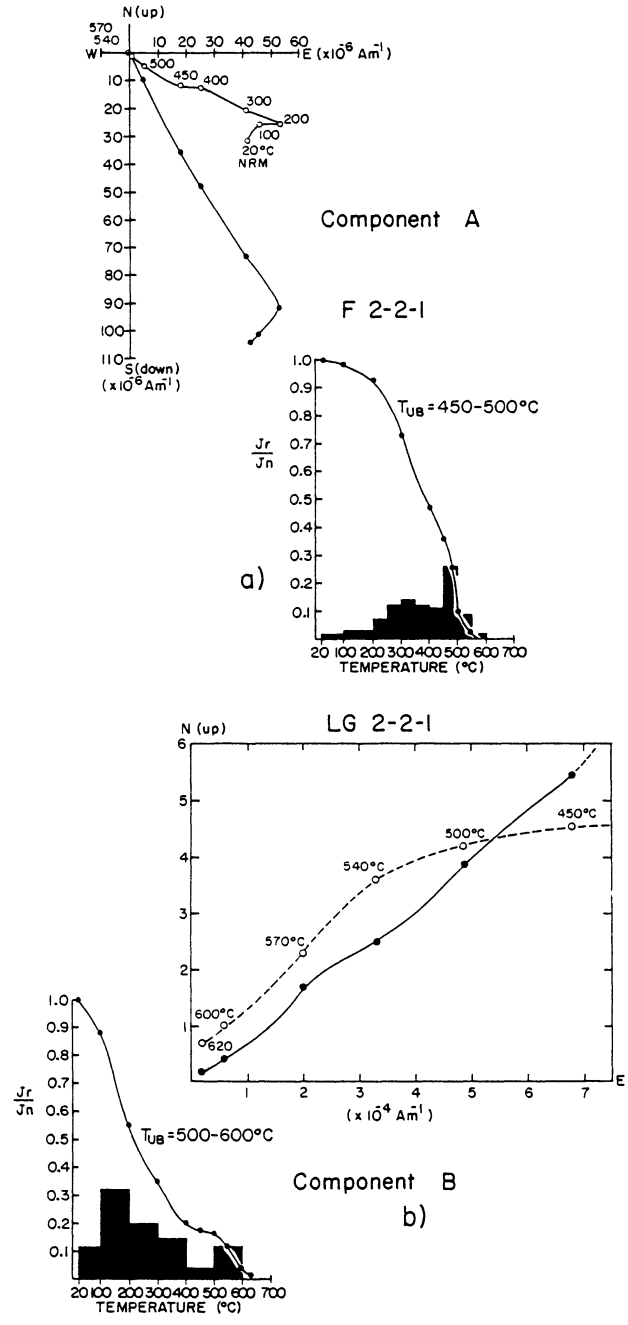


Fig. 4. Representative orthogonal thermal demagnetization diagram (ZIJDERVELD, 1967) of successive end-points of magnetization vector for component A (specimen F2-2-1), component B (specimen LG2-2-1), component C (specimen E2-5-2) and component D (specimen P2-5-1). The solid (open) circles represent projections in the horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps in (°C). Representative normalized intensity (J_r/J_n) decay curves and unblocking temperatures for the same specimen in the bottom part of the diagrams.

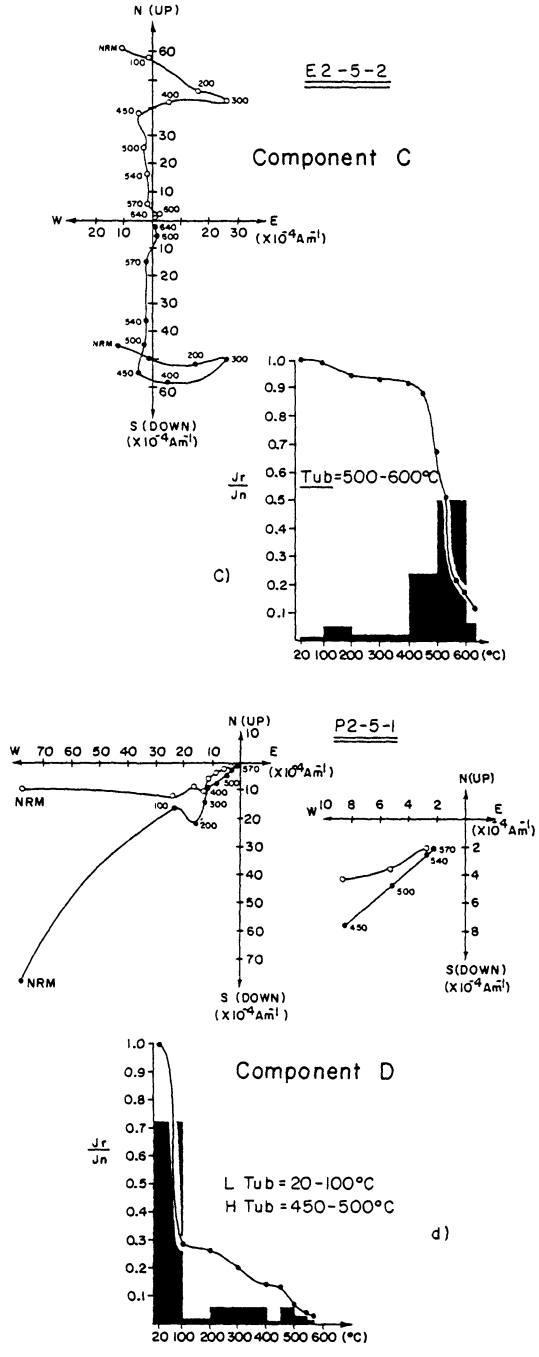


Fig. 4. (continued).

Table 1. Structural data at the sampling sites.

Site No.	N	n	Strike (°)	Dip (°)	in situ NRM D (°)	I (°)	α_{95} (°)	k
E-1	5	12	—	1	19	60	39	5
E-2	5	13	345	7	16	76	43	4
F-1	5	12	328	6	282	17	51	3
F-2	5	16	320	16	136	81	39	5
F-3	5	11	281	17	331	25	42	4
P-1	5	14	241	6	89	24	51	3
P-2	6	17	82	75	183	35	40	5
LG-1	5	14	143	10	338	60	33	6
LG-2	5	15	82	47	325	74	31	7

N: number of samples, n: number of specimens.

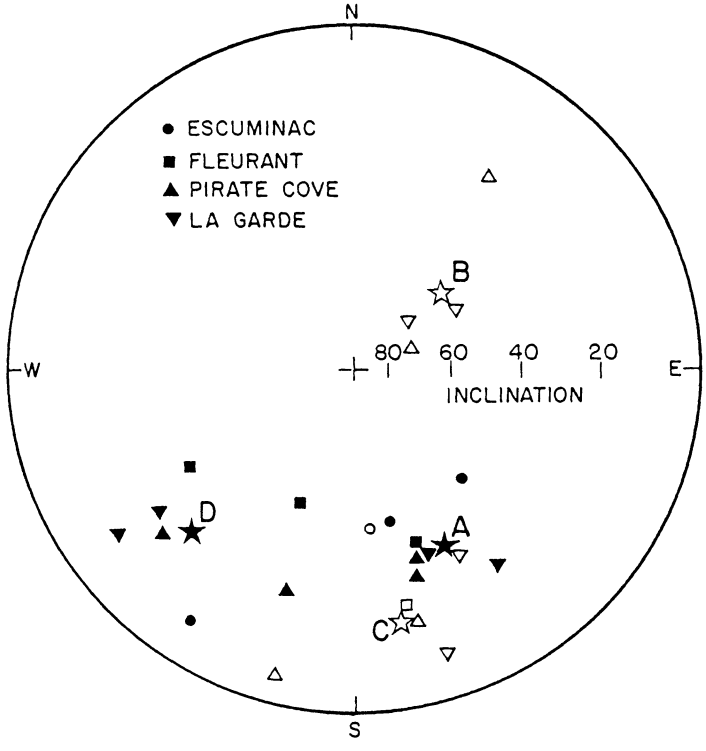


Fig. 5. Equal-area projections of site mean directions of magnetization (components A, B, C and D) obtained after AF and thermal treatment without tilt correction (see Table 2).

Table 2. Site-mean directions of magnetization after AF and thermal demagnetization.

Site	N	n	In situ				Tilt-corrected				P
			D (°)	I (°)	α_{95} (°)	k	D	I	α_{95} (°)	k	
Component A											
E-1	2	5	135	44	—	—	135	44			R
E-2	3	6	168	42	28	14	162	45			R
F-2	3	10	160	35	33	12	150	45			R
P-1	3	9	162	30	42	9	163	36			R
P-2	5	16	164	25	20	15	161	− 49			M
LG-1	3	9	158	30	31	11	162	23			R
LG-2	2	7	143	20	—	—	147	− 27			M
			156	31	10	34	154	19	30	5	
Component B											
P-1	3	5	34	− 21	27	22	34	− 22			M
P-2	2	4	71	− 77	—	—	7	− 12			M
LG-1	2	4	60	− 54	—	—	58	− 44			M
LG-2	2	5	46	− 67	22	20	13	− 27			M
			52	− 55	28	13	26	− 28	27	12	
Component C											
E-2	4	7	171	− 42	21	20	176	− 38			R
F-3	2	5	167	− 20	—	—	167	− 3			R
P-1	4	9	165	− 17	21	20	164	− 12			R
P-2	3	10	196	− 5	5	636	224	− 40			R
LG-1	1	3	146	− 27	—	—	140	− 31			R
LG-2	2	4	160	− 9	—	—	169	− 54			R
			167	− 21	17	16	171	− 32	26	8	
Component D											
E-2	2	6	213	9	—	—	212	14			R
F-2	3	9	202	47	30	18	200	61			R
F-3	2	6	240	33	—	—	252	36			R
P-1	2	5	230	19	—	—	232	18			R
P-2	2	7	197	24	—	—	196	− 20			M
LG-1	1	2	236	12			239	6			R
LG-2	3	5	231	20			236	3			R
			223	25	18	12	225	18	25	6.5	

P: polarity, N: normal, R: reverse, M: mixed.

situ component B would be located in the Early Devonian segment of the APWP curve. This observation does not make sense for two reasons: 1) the two formations containing the B component are mainly of early Middle Devonian and not Early Devonian age and, 2) a secondary component cannot have an older age than a primary

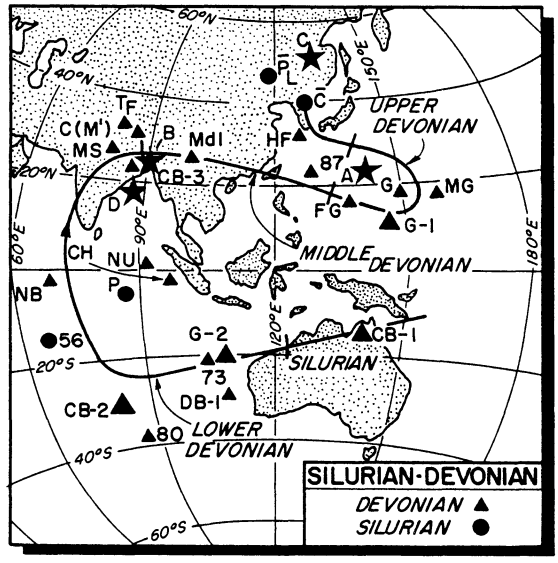


Fig. 6. Pole positions of the four components (A, B, C and D) indicated with a star on the proposed Devonian APWP of cratonic North America (SEGUIN, 1986). The other poles are listed in ROY *et al.*, (1983). The additional poles include: CH (SEGUIN *et al.*, 1982b), DB-1 (MURTHY, 1983), G-1 and G-2 (SEGUIN, 1986), TF (SPARIOSU and KENT, 1983) and NB (SEGUIN and GAHÉ, 1985). The supporting poles are considered as the most reliable for this timespan.

component. In conclusion, the arguments for a primary age of component B are more numerous than for a secondary age. Thus components A, C and D are apparently secondary while component B may be primary or secondary but more certainly primary. The conglomerate test is applicable in the case of the Fleurant and La Garde formations. In the Fleurant Formation, the pebbles (boulders) sampled include one limestone but mainly sandstones and arkoses; these boulders are weakly magnetized. The pebbles sampled in the La Garde Formation are composed mainly of strongly magnetic hard volcanic rocks. The cleaned directions of the pebbles in the La Garde Formation (samples L1.4 and L1.5) coincide with those of the A and D components implying that this test is clearly negative. Indeed, both the pebbles and the matrix were remagnetized otherwise they would not have the same magnetic directions. In the Fleurant Formation, some of the cleaned directions in the pebbles (samples F1.4 and F1.5) also coincide with the directions of components A and D, but other pebbles (samples F3.1 and F3.2) do not, which means that they were not remagnetized. Pending on their magnetic mineralogy, some pebbles were remagnetized while others were not. Consequently in this formation, the conglomerate test is definitely negative for components A and D but this needs not be the case for components B and C. On the basis of the conglomerate test, it is then concluded that components A and D are definitely secondary while component B may be primary. Our interpretation of the

conglomerate test is that component B was not recorded in the boulders of the Fleurant Formation simply because the boulders were not remagnetized at the time of deposition of the Fleurant Formation.

The final conclusion is that based on both the fold and conglomerate tests; components A, C and D are interpreted as secondary while component B is most probably primary. We will see in the next section that the second phase of the Acadian orogeny occurred at the end of the deposition of the Pirate Cove Formation. This observation is in agreement with the occurrence of the B component in the La Garde and Pirate Cove formations only. Component C is a late chemical magnetization which affected mainly the matrix of the rock units but very little the grains and the boulders.

9. Paleomagnetic Poles and Their Age: A Discussion

Based on group mean remanence directions, paleomagnetic poles are calculated for each group, (A, B, C and D) (Fig. 5, Table 2). Assuming that component B is primary, the age of the tilt-corrected B component is probably early-Middle Devonian. If component B is secondary, it is most likely related to the second phase of the Acadian orogeny since it is not found in the overlying Fleurant and Escuminac formations; consequently, its age of acquisition is late Middle Devonian. Assuming that component B was acquired when the bedding plane was horizontal, the tilt correction is valid to restore the paleomagnetic direction and to estimate the paleopole.

After taking into account the relative ages of the isolated components discussed in the previous section, the paleopoles for the in situ and tilt corrected B component as well as in situ A, C and D components are retained in this section. The paleopole of the in situ B component (in its normal polarity) is 75° E 6° S (dp=23°, dm=35°). The paleopole position of the tilt corrected B component (in its normal polarity) is 89° E, 23° N (dp=17°, dm=30°) (Fig. 6). The significant conglomerate test strongly suggests that the paleopole position of the tilt corrected B component is the one to be retained. For the secondary components (A, C and D), the in situ paleomagnetic field direction is used instead of the tilt corrected one because it is certain that phase B of the Acadian orogeny took place in the interval of time located between the deposition of the Pirate Cove Formation and the Miguasha Group. As the Fleurant and Escuminac formations were only slightly tilted afterwards, it is evident that the detection of similar secondary components in all four geological formations imply that they were acquired after the last tilting motions and consequently structural corrections are not warranted for these components. The paleopole positions of the in situ A, C and D components (in their normal polarity) are: 141° E, 22° N (dp=6°, dm=11°); 136° E, 51° N (dp=9°, dm=18°) and 83° E, 17° N (dp=10°, dm=19°) respectively.

The paleopole position of component C falls on the Permian pole position (Fig. 6) for cratonic NA (Kiaman interval). Thus, hot and humid climate in a terrestrial environment is responsible for the chemical origin of this remanence component. The paleopole position of component A (Fig. 6) falls on the mid-Late to very Late

Devonian sector for cratonic NA (SEGUIN, 1986). This suggests that component A is an overprint related to the Acadian orogeny. DINELEY and WILLIAMS (1968b) have shown that the Acadian orogeny is a complex of Earth movements occurring as a pulsatory sequence of uplifts. These authors have demonstrated that three phases of Acadian orogeny can be identified: Acadian A, B and C. Phase Acadian A which is pre-La Garde Formation represents the major disturbance. It occurred in Early Devonian time and it metamorphosed and disrupted all previously existing formations. Phase Acadian B took place at the end of Pirate Cove time; it is pre Miguasha-Group and occurred in Middle to Late Devonian time. The movements accompanying this phase had smaller disturbance effects than phase Acadian A. Phase Acadian C is post-Escuminac and pre-Bonaventure (probably Famennian); its disturbance effect is very minor. On this basis, component A is probably closer to very Late Devonian than mid-Late Devonian. The Alleghenian (Hercynian) orogeny was absent in this region; the Carboniferous sedimentary rocks (Bonaventure Formation) are subhorizontal.

The paleopole position of component D is located close to the pole of the tilt-corrected component B (Fig. 6). Component D is tentatively assigned a late-Middle or early-Late Devonian age. This age bracket for the acquisition of magnetization of component D is constrained by the fact that this component is secondary and thus younger than component B (early-Middle Devonian). In addition, component D is present in all four geological formations; consequently, it is younger than the second phase (B) of the Acadian orogeny (late-Middle Devonian). It is most probably older than component A (mid-Late Devonian). The highest degree of alteration (e.g., surface of the cores) and hematitization are found in the cylindrical specimens containing mainly components A and C. This observation also suggests that components A and C are the youngest, and D and B the oldest. The pole of component (D) should then be positioned in the late-Middle and mid-Late Devonian timespan, i.e., between poles B and A on the APWP of North America (Fig. 6). This is not quite the case, pole D is located slightly to the South of B and especially too far west. Consequently, the relationship of component D with the other components is still not well understood. One can hypothesize that this component is an unreliable paleomagnetic direction or that it is the result of composite magnetizations or else that the Devonian APWP of North America is more complicated than the one presented in Fig. 6.

The paleomagnetic results of this study are brought into focus by comparing them with other results of the same age originating from cratonic North America (NA) and the Appalachians (A). On cratonic NA, the geological formation which was best studied paleomagnetically is the Upper Devonian Catskill redbeds (KENT and OPDYKE, 1978). Unfortunately, its magnetization is probably a Permian overprint; its paleopole position is 117° E, 47° N. This last value is reasonably close to the paleopole position of component C. In the northern Appalachians, the most important and/or recent paleomagnetic studies include the Perry Formation (ROBERTSON *et al.*, 1968), the Terrenceville Formation (KENT, 1982), the Compton Formation (SEGUIN *et al.*, 1982a), the Hersey and Eastport formations (KENT and OPDYKE, 1980), the Traveler felsite (SPARIOSU and KENT, 1983), the Gaspé sediments (SEGUIN, 1986) and the New

Brunswick volcanics (SEGUIN and GAHÉ, 1985). The two first rock units are Upper Devonian and the others Lower Devonian. The respective paleopole positions are: 118° E 32° N, 124° E 27° N, 79° E 20° N, 122° E 22° N, 82° E 29° N, 109° E 20° S and 88° E 1° N. The magnetization of the Perry as well as Hersey and Eastport formations may well be secondary (ROY, 1982; KENT and OPDYKE, 1982). According to STEARNS and VAN DER VOO (1987), the Perry Formation was remagnetized in the Early Carboniferous time. In the central Appalachians, only the Upper Devonian Catskill Formation (MILLER and KENT, 1986a, b) yielded for the first time a primary magnetization; its paleopole position is 124° E 26° N.

Still, it is quite clear that the longitudes of the Lower Devonian paleopoles are located further to the east than those for the Upper Devonian units and that latitudes of the poles are more variable for the Lower Devonian units. A comparison of the paleopole of component A with the above mentioned studies shows clearly that its poles are located close to the Upper Devonian Terrenceville pole. The paleopole of component B is much closer to the Lower Devonian poles of Maine, New Brunswick and Quebec. The paleopole of component D should be located some 30° further east; this situation remains unexplained.

10. Conclusion

New paleomagnetic results from Devonian formations of the northern Appalachians of North America are presented. After AF and thermal demagnetizations, four components of magnetization are obtained. Application of the fold and conglomerate tests suggests that three of the components are secondary while one may be primary or secondary. Additional evidence supporting this assertion are based on the degree of alteration of the magnetic minerals and the respective distribution of the four components of magnetization in the upper-Lower to Upper Devonian formations. One component (C) is interpreted as having a Permian age (Kiaman interval). It is mainly found in the more hematitized La Garde and Pirate Cove formations; it represents clearly the latest event of magnetization. The paleopole of component A is situated slightly to the south of Early Carboniferous poles from New Brunswick (SEGUIN *et al.*, 1985; SEGUIN and FYFFE, 1986). Consequently, component A is most likely of mid- to very Late Devonian age and component D of late-Middle to mid-Late Devonian age. The pole position of component B (89° E, 23° N) is located very close to the late-Early to early-Middle Devonian Compton Formation of the Quebec Appalachians (SEGUIN *et al.*, 1982a) which is interpreted as primary. There is thus a fair probability that component B represents a primary component of magnetization characteristic of the late-Early to early-Middle Devonian APWP for the North American continent (Fig. 6). The velocity of NA with respect to the pole is of the order of 3° Ma^{-1}, i.e. 10^{-3} s yr^{-1}. This last value is relatively large but certainly not unrealistic.

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